Hydration Guidelines Before InBody BIA Testing

Adequate and consistent hydration is important for Bioelectrical Impedance Analysis (BIA), as the measurement is influenced by the distribution of body water. For reliable and comparable InBody results, it is recommended to maintain **normal hydration** and avoid both dehydration and excessive fluid intake immediately before the test.

Recommended guidelines:

- Maintain your **usual hydration pattern** before the InBody test.
- Avoid drinking a **large quantity of water immediately before the measurement**.
- If water has been consumed shortly before the test, it is recommended to **allow at least 20 minutes before taking the InBody measurement**.
- Avoid testing immediately after strenuous exercise, as exercise can temporarily alter fluid distribution.
- For repeated measurements, try to maintain **similar testing conditions**, preferably at a similar time of day and under similar food, fluid and activity conditions.
- When possible, testing in a **fasted or standardized state** can help improve consistency between measurements.

These recommendations are intended to improve **measurement standardization and comparability**, particularly when InBody measurements are being used to monitor changes over time.

Scientific Basis

Research indicates that acute fluid intake can influence BIA-derived parameters, including intracellular water, extracellular water, skeletal muscle mass and fat mass. Therefore, standardizing hydration and pre-measurement conditions is important when interpreting BIA results.

The ESPEN guidelines for BIA also recommend standardizing factors such as body position, previous exercise and dietary intake, noting that food and beverage consumption can influence impedance measurements.

References

1. Kyle, U. G., Bosaeus, I., De Lorenzo, A. D., Deurenberg, P., Elia, M., Gómez, J. M., Lilienthal Heitmann, B., Kent-Smith, L., Melchior, J.-C., Pirlich, M., Scharfetter, H., Schols, A. M. W. J., & Pichard, C. (2004). ESPEN guidelines for bioelectrical impedance analysis (part II: Utilization in clinical practice). *Clinical Nutrition*, *23*(6), 1430–1453. https://doi.org/10.1016/j.clnu.2004.09.012
2. Schierbauer, J., Günther, S., Haupt, S., Zimmer, R. T., Herz, D., Voit, T., Zimmermann, P., Wachsmuth, N. B., Aberer, F., & Moser, O. (2023). Acute fluid intake impacts

assessment of body composition via bioelectrical impedance analysis: A randomized, controlled crossover pilot trial. *Metabolites, 13*(4), 473. https://doi.org/10.3390/metabo13040473

3. Kerr, A., Slater, G. J., & Byrne, N. (2017). Impact of food and fluid intake on technical and biological measurement error in body composition assessment methods in athletes. *British Journal of Nutrition, 117*(4), 591–601. https://doi.org/10.1017/S0007114517000551
4. Herberts, T., Slater, G. J., Farley, A., Hogarth, L., Areta, J. L., Paulsen, G., & Garthe, I. (2023). Protocol standardization may improve precision error of InBody 720 body composition analysis. *International Journal of Sport Nutrition and Exercise Metabolism, 33*(4), 222–229. https://doi.org/10.1123/ijsnem.2022-0219